



BSc (Hons) in **Information Technology**
Specialized in AI

IT3012 : Intelligent Agents

Year 3 · Semester 1 · 2026 Semester 2

Faculty of Computing

Practical 02

Objective & Lecture Mapping: In Lecture 03, we learned that a mathematically perfect "Table-Driven Agent" is impossible to build due to combinatorial explosion. Instead, we must write Agent Programs. Today, you will implement a **Simple Reflex Agent** using Condition-Action rules, observe its fatal flaws in a partially observable environment, and then upgrade it to a **Model-Based Agent** that maintains an internal memory state.

Part 1: Practical Implementation — Coding the Architectures

Step 1.1: Setting the Trap (Partial Observability)

To see why Simple Reflex agents fail, we must handicap your agent's sensors. We are moving from a Fully Observable world to a Partially Observable one.

1. Open your `visual_grid_game.py` from Lab 01.
2. Locate the `get_percept(self)` method.
3. Modify it so it **no longer returns** the exact global coordinates (`agent_pos`). Instead, it should only return local booleans, e.g., `{ 'wall_ahead': True/False, 'food_here': True/False }` based on checking the adjacent cell in its current facing direction.

Step 1.2: The Simple Reflex Agent (Implementation & Failure)

1. Create a new class called `SimpleReflexAgent` .
2. Implement a `sense_and_act(self, percept)` method that uses strictly IF-THEN logic (Condition-Action rules). *Do not use an `__init__` method to store history.*

Example Logic: IF food_here THEN suck; IF wall_ahead THEN turn_left; ELSE move_forward

3. Run the simulation. **Observe:** Watch the agent get trapped in a corner or a U-shaped wall, infinitely repeating a cycle (e.g., turn left, step forward, hit wall, turn left...).

Step 1.3: The Model-Based Agent (Memory & State)

1. Create a new class called `ModelBasedAgent`.
2. Implement an `__init__(self)` method to initialize an internal memory state (e.g., `self.visited_cells = set()` or a relative movement tracker).
3. In `sense_and_act(self, percept)`, first **update the state** (Transition & Sensor Model) by recording the current percept and the last action taken.
4. Update your IF-THEN rules to query the memory.
Example Logic: IF wall_ahead AND left_is_visited THEN turn_right
5. Run the simulation. **Observe:** The agent should now remember where it has been, realize it is in a loop, and choose an alternate path to escape.

Part 2: Theoretical Evaluation & Lecture Mapping

Based on your practical observations above, answer the following questions to map your code directly to the architectural theories covered in Lecture 02.

1. (Remember) According to Lecture 02, why is it impossible to program a mathematically perfect "Table-Driven Agent" for complex environments like Chess? What happens as the agent's lifetime increases?

It is impossible because a Table-Driven Agent would need to store an action for every possible percept sequence. In a complex environment like Chess, the number of possible percept sequences is extremely large. As the agent's lifetime increases, the percept sequence becomes longer and the number of possible sequences grows enormously, making the table impractical to store and use.

2. (Understand) Look at the code you wrote for your `SimpleReflexAgent` . Identify and explain the specific lines of code that represent the "Condition-Action Rules" discussed in the lecture.

IF food is ahead THEN move right.
IF wall is ahead THEN move left.
No food and no wall ahead. Action: Move right.

3. (Analyze) Your `SimpleReflexAgent` likely got stuck in an infinite loop during Step 1.2. Based on the lecture, analyze exactly why this happened. How did the combination of "Partial Observability" and a lack of "Percept History" cause this failure?

The `SimpleReflexAgent` can get stuck in an infinite loop because it only uses the current percept and does not maintain percept history. Due to partial observability, the agent receives only limited information about the environment, such as whether there is a wall, food, or toxin ahead. It cannot remember previously visited cells or actions. Therefore, when it encounters the same situation again, it applies the same condition-action rule again, which can cause it to repeatedly move through the same cells and become stuck in an infinite loop.

4. (Evaluate) In Step 1.3, you added an internal state to your `ModelBasedAgent` . Evaluate how your specific code handles the "Transition Model" (how the world evolves) and the "Sensor Model" (how the agent's actions affect the world).

My `ModelBasedAgent` has a basic internal state, but the Transition Model is only partially implemented. The agent stores the last action and previously observed states using `self.last_action` and `self.visited_cells`. However, it does not explicitly predict how an action will change the agent's position; the environment's `execute_action()` method handles the actual state transition. The Sensor Model is implemented through `get_percept()`, which provides information such as whether there is a wall, food, or toxin ahead. The agent uses these percepts to select its next action. Therefore, the agent has a basic Sensor Model and a limited Transition Model.

Finalize and Lock Lab 02 Documentation



FACULTY OF COMPUTING